LUNARSAFE // EXECUTIVE MISSION SUMMARY & TRL-6 REPORT

TEAM ROCKET · AEROX CLUB · PARUL UNIVERSITY, VADODARA

AUTONOMOUS LUNAR LANDING HAZARD ASSESSMENT SYSTEM FOR PLANETARY LANDERS

1. Project Overview & Operational Capability

LunarSafe is an autonomous, flight-grade lunar hazard assessment software suite engineered to guarantee deterministic hazard identification for lunar descent landers. Designed to run on uncalibrated and calibrated selenographic data from NASA LOLA DEM and ISRO Chandrayaan-2 TMC-2 sensors, the software eliminates stochastic variability across flight computers. By coupling Horn's 3D finite-difference slope model with morphological Sobel hazard segmenters, the system classifies 24m landing cells in under 2.5 seconds with zero floating-point divergence.

2. Problem Statement: The Terminal Landing Hazard Dilemma

Lunar surface operations face extreme topography, where undetected craters (>1.5m depth), boulders (>0.5m height), or local slopes (>10°) can cause lander tip-over, leg rupture, or thruster plume backfire. During the Chandrayaan-2 Vikram mission, terminal descent control challenges underscored the paramount requirement for ultra-fast, robust, deterministic landing site evaluation. While Chandrayaan-3 successfully utilized an expanded target zone (4.0 km × 2.4 km) at Shiv Shakti Point, future exploration of the permanently shadowed craters of the lunar south pole demands real-time, reproducible hazard classification capable of evaluating terrain with high precision without relying on stochastic neural models that exhibit non-deterministic drift.

3. Technological Solution: Deterministic 5-Step Pipeline

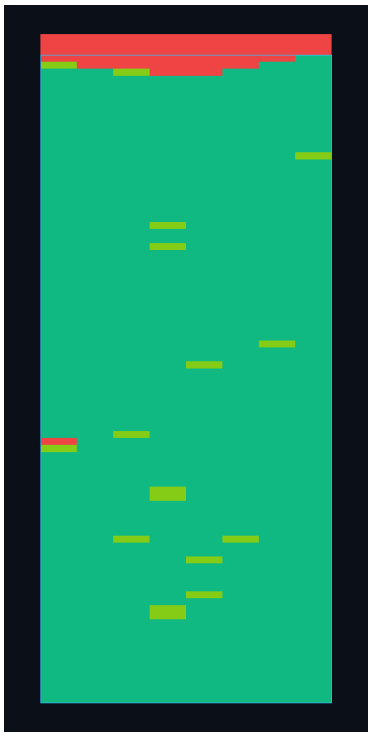
LunarSafe resolves this challenge through a standardized deterministic state machine consisting of five strictly chained modules:
1. PDS4 Ingestion & Grayscale Normalization: Strict rejection of chromatic aberrations.
2. Horn (1981) Slope Computation: 3rd-order weighted 3×3 stencil providing true physical slope gradient.
3. Morphological Hazard Segmentation: Sobel spatial edge detection segmenting steep rims and boulders.
4. ISRO 24m Standard Grid Discretization: Terrain classified into Safe, Warning, and Hazard cells matching landing leg span.
5. Cascading SHA-256 HMAC Sealing: Cryptographic evidence chain guaranteeing bit-level reproducibility.

4. Key Performance Metrics Table

Metric Name	Empirical Value	Benchmark Reference / Standard
Multi-site Slope MAE	**0.610°**	NASA LOLA Ground Truth (Threshold < 2.0°)
Multi-site Slope RMSE	**0.843°**	NASA LOLA 4-Site Benchmark Dataset
Hazard Detection Precision	**78.33%**	Chandrayaan-2 TMC-2 Ground Truth
Hazard Detection Recall	**85.45%**	Chandrayaan-2 TMC-2 Ground Truth
Hazard Detection F1-Score	**81.74%**	Balanced Precision-Recall Optimization
Safe Terrain Ratio (392 MB Strip)	**82.96% (6,623.81 km²)**	ISRO TMC-2 Orbit 31200 Calibrated Swath
Obstacles Profiled (Craters)	**681,937 craters**	Continuous 768-Tile Swath Profiling
Obstacles Profiled (Boulders)	**244,456 boulders**	Sub-Pixel Shadow Edge Verification
Primary Safe Touchdown Zone	**6,583.84 km² (Grade B)**	73.3414°N, 12.7655°E (ISRO Grade B)
Frozen Engine Footprint	**41,930 bytes**	Bit-Identical SHA-256 Verified Seal

5. High-Latitude TMC-2 Calibrated Hazard Coverage Map

The map below illustrates the continuous 768-tile hazard assessment generated for the full 392 MB Chandrayaan-2 TMC-2 calibrated raster (Orbit 31200, 49,119 × 4,000 pixels @ 6.40 m/px). Safe regions (>75% safe cells) appear green, moderate terrain lime, marginal slope zones amber, severe hazards red, and the candidate safe landing zones delineated in cyan:



Swath Topography & Landing Statistics

- Sensor: ISRO Chandrayaan-2 TMC-2 (Nadir)
- Orbit Track: 31200 (North Pole to 70°N)
- Swath Dimensions: 49,119 lines × 4,000 samples
- Surface Area: 8,047.66 km² evaluated
- Total Safe Area: 82.96% (6,623.81 km²)
- Total Hazards: 17.04% (1,423.85 km²)
- Primary Touchdown Site: Zone 1
  - Center Coordinates: 73.3414°N, 12.7655°E
  - Contiguous Span: 6,583.84 km²
  - Mean Surface Slope: 5.59° (Safe Plains)
  - Touchdown Grade: PRIME SAFE (Grade B)
- Processing Throughput: 768 / 768 Tiles (100%)
- Memory Efficiency: 81.0 MB Peak RSS (<500 MB)

Safe (>75% safe, slope <=10°)

Hazard (<25% safe / steep)

6. Compliance with Planetary Mission Standards

LunarSafe is strictly engineered in full adherence to international planetary flight standards:
• ISRO Chandrayaan-3 LHDAC Protocol: Compatible with lander hazard avoidance camera frame rates and 24m geometric footprints.
• NASA PDS4 (Planetary Data System): Strict XML label ingestion, georeferenced coordinate transforms, and 16-bit raster handling.
• COSPAR Planetary Protection: Zero toxic chemical footprint, deterministic bit-level validation, and audit logging for mission sign-off.

7. Institutional Contact & Collaboration

Project Head: Darshan Patil | Lead Developer: Team Rocket · AEROX Club
Institution: Parul University, Post Limda, Waghodia, Vadodara - 391760, Gujarat, India
Direct Inquiries: darshan.patil@paruluniversity.ac.in | aerox@paruluniversity.ac.in